

law. Large scale search systems for the purpose of information exchange between Internet communities leads to the formation of covert channels. An agent-based security model to control data from covert channels can help solve the problem of data leakage in the cloud environment.

Data loss

The very prospect of losing one's data permanently is terrifying. Even in the cloud, data loss is a major concern. Losses could be caused by malicious attackers, accidental deletion by cloud service providers, or physical catastrophes like fire or earthquakes. Another possibility is if the customer forgets the encryption key of the data encrypted; this could also lead to permanent loss of data. Once the data is lost, it is forever irretrievable unless the cloud services provider maintains a backup for the data at all times, though this could indirectly make data breaches easier.

Account or service traffic hijacking

Account or service traffic hijacking could happen through various attack methods such as phishing (the attempt to acquire sensitive information such as usernames, passwords and credit card details by masquerading as a trustworthy entity in an electronic communication), fraud and exploitation of software vulnerabilities. If attackers gain access to your credentials, they can eavesdrop on your activities and transactions, manipulate data, return falsified information, and redirect your clients to illegitimate sites. Your account or service instances may become a new base for the attackers, who may leverage the power of your reputation to launch subsequent attacks. With stolen credentials, attackers can also access critical areas of deployed cloud computing services, compromising the confidentiality, integrity and availability of those services.

In April 2010, Amazon experienced a Cross-Site Scripting (XSS) bug that allowed attackers to hijack data from the site. How do you eliminate the risk of your data being stolen? Easy! Don't trust anyone. Your encryption keys should always be under your own control. Do not allow your cloud services provider to control your keys. Another important factor to consider here is not to only avoid such hijacking, but also have solutions for mitigating the damage, in case an attack occurs. One way to do this is to segment your encryption. Each encryption project can contain as much (or as little) data as required, across multiple disks, databases, file servers and object storage. Organisations should look at prohibiting the sharing of account credentials between users and service providers, and leverage strong two-factor authentication techniques wherever possible.

Insecure interfaces and APIs

There are several Application Programmer Interfaces (APIs) used by cloud service managers to provide services to the users. Provisioning, management, orchestration and monitoring are all performed using these interfaces. The security of these APIs determines the security and availability of general cloud

services. Any insecurity in the interfaces or APIs could cause a catastrophe in cloud services. From authentication and access control to encryption and activity monitoring, these interfaces must be designed to protect against both accidental and malicious attacks. Also, organisations and third parties often build upon these interfaces to offer value-added services to their customers. This introduces the complexity of the new layered API; it also increases risk, as organisations may be required to relinquish their credentials to third-parties in order to enable them to operate. The organisation could fall prey to the various security issues related to confidentiality, integrity, availability and accountability if it relies upon a weak set of interfaces and APIs.

The APIs key should be protected but unfortunately, there is not too much awareness on how to protect these keys. Cloud and Web service developers must first follow certain best practices in opening up their APIs and interfaces to third parties and, in turn, third-party developers need to handle these in a secure manner. The developers also need to understand the limitations and the best practices around implementing API keys and interfaces.

Denial of Service attacks

As the name suggests, this simply means denying users the cloud services, to the extent that they might not even be able to access their own information. Denial of Service (DoS) attacks are caused by forcing the victim cloud service to not respond, thus leaving the users angry. This is mainly caused by making the cloud service consume large amounts of resources like memory, processor power, disk space or network bandwidth. The attacker (or attackers, in the case of distributed denial of service attacks) causes an intolerable system slowdown thus leaving the users confused and unhappy. Asymmetric application-level DoS attacks take advantage of vulnerabilities in Web servers, databases or other cloud resources, allowing a malicious individual to take out an application using an extremely small attack payload (which could be less than 100 bytes).

Experiencing a DoS attack is like being caught in a traffic gridlock; there is no way to reach your destination and nothing else to be done other than to sit and wait patiently. Using an Intrusion Detection System (IDS) is the most popular defence against DoS attacks. Each cloud is loaded with a separate IDS. The different IDSs work on the basis of information exchange. In case a specific cloud is under attack, the co-operative IDS alerts the whole system. A decision on the trustworthiness of a cloud is taken by voting and the overall system performance is not hampered.

Malicious insiders

Malicious insiders within an organisation can cause the whole cloud service to become unstable. These people could include malicious administrators who hold the key to managing all the data, files and privileged company resources; tech-savvy individuals who can use their knowledge about the weaknesses of a company's security to breach clearance and access